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Executive Summary

【82†embed_image】Marginal gains theory—making many small, 1 % improvements—can transform complex engineered systems like a microprocessor into high performers【25†L37-L42】【66†L175-L183】. In semiconductor design and manufacture, this means iterating on architecture, EDA flows, fab processes, test and packaging to scrape out every percent of gain. The literature and industry case studies confirm that incremental optimisation (e.g. Design-for-Yield, tighter timing, improved process control) yields substantial aggregate benefits【34†L41-L47】【70†L269-L273】. We review academic and whitepaper sources, analyse examples (node shrinks, yield ramp, DFT coverage, etc.), and map "marginal-gain" tactics to each stage of SilicaFlux's lifecycle. A decision matrix ranks ~12 candidate tweaks (by effort, cost, risk, etc.) and we propose a phased SilicaFlux plan with KPIs for engineering samples, pilot lots and volume. Finally, we list required tools/data, outline experiments (noting any proprietary blockers), and give CI metrics and timeline charts. Key sources include foundry/EDA documentation, SIA reports and peer-reviewed papers.

1. Marginal Gains Theory in Engineering

Definition: Marginal gains (or continuous improvement) is the philosophy of accumulating many small improvements to achieve a large overall benefit【25†L37-L42】【66†L175-L183】. Popularised by Coach Dave Brailsford's British Cycling team, the idea is to "break down everything you could think of … and then improve it by 1 %" so that the tiny gains multiply into a big performance jump【25†L37-L42】. In complex engineered systems, which have many interacting parts, focusing on many minor optimisations often beats waiting for a single breakthrough.

Relevance to engineered systems: In manufacturing and software engineering, this mirrors *continual improvement* (Kaizen) principles【66†L129-L132】【66†L175-L183】. For example, quality-management standards advocate iterative Plan–Do–Check–Act cycles. In semiconductor design, where chips contain billions of transistors and intricate interconnect, there are **countless** opportunities to tune power, timing, layout, test coverage, etc. Small enhancements in logic synthesis, placement, process steps or materials can yield compounded performance, yield or cost gains. As one source notes, a continual-improvement mindset "starts with the customer to create needed value with fewer resources and less waste"【66†L129-L132】. In sum, marginal gains theory justifies a methodical, cross-disciplinary effort to eke out percent-level improvements across RTL, EDA tools, fabrication, and supply-chain processes.

2. Survey of Incremental Optimisation in Semiconductors

2.1 Design and EDA Flows

Academic and industry literature repeatedly show that *small design and tool optimisations* improve chip metrics. For instance, research on **incremental physical design** (ECO, routing guides) has demonstrated major speedups: one study found an incremental gate-sizing update ran *~283× faster* than a full re-synthesis, for the same final timing【70†L269-L273】. Another example used an "incremental floorplanner" to rewire interconnect while preserving area; this reduced total wirelength (and thus delay) without changing chip area【70†L212-L218】. In practice, design teams apply many 1–5 % optimisations: retiming critical paths, resizing buffers, enabling clock gating, or hand-injecting

multi-corner constraints. Vendors note that each new tool release or P&R run often yields a few percent better PPA (performance/power/area).

Design-for-Yield (DFY): Literature on DFY techniques emphasizes improving yields by design rules and layout choices. An EE Times article reports that **"measurable results"** show DFY methods *"overwhelmingly demonstrate that design yield can be improved"* through careful patterning, redundancy and variation-tolerant circuits【73†L184-L188】. This includes, for example, inserting spare vias or cells, avoiding minimum-spacing hotspots, and optimizing analog layouts for matching. Academic groups (CMU, Edinburgh) and large firms have built specialized tools to plug DFY gaps【73†L251-L259】. Such efforts typically shave fractional percentages off defect sensitivity (e.g. particle short/open risk), but in aggregate they reduce wafer scrap significantly.

EDA Flow: Modern flows also automate incremental loops. For instance, congestion analysis and back-annotation can adjust routing in small steps. One paper describes an "incremental placement" tool (FREe) that quickly reroutes nets to even out congestion with no full optimization loop【70†L236-L244】. Likewise, engineers iteratively tighten timing constraints or split critical regions into multiple engine runs. There are also continuous improvements in synthesis (logic restructuring) and STA engines. Where academia highlights "incremental design methods" as crucial for faster time-to-market (e.g. handling ECOs efficiently)【70†L261-L270】, industry blogs and white papers from Cadence/Synopsys often stress that applying each new PPA tweak or even re-using proven IP blocks yields reliable small gains.

2.2 Process Engineering and Yield

Yield optimization is a classic area of marginal gains. The Semiconductor Industry Association observes that **"yield is the single most important factor in overall wafer processing costs, as incremental increases in yield significantly reduce manufacturing costs"**【34†L41-L47】. In practice, fabs run *Statistical Process Control (SPC)* and Advanced Process Control (APC) loops continuously, tweaking deposition, etch times, temperature or beam focus to squeeze out yield. McKinsey's industry report exemplifies this: simple equipment calibrations and material changes reduced certain defect rejects by *tens of percent* (e.g. contaminants –90%, wrinkles –40%)【87†L661-L664】. These are the kind of "small wins": adjusting lamination material (9% gain), fine-tuning bake processes (40% gain), or running defect analytics【87†L669-L678】. Over many generations, yields ramp gradually as defects and variability are chipped away by these local fixes.

Academic fab-research similarly endorses continuous improvement. For example, adjustments in mask corner exposure or chemical concentrations may each improve line-edge roughness slightly, but together enable a new node shrink. As Moore's Law progresses, the literature notes that "risks to yield due to process variability and contamination are ever increasing, as is the importance of continuously improving design and machine capabilities"【34†L62-L69】. Fabs therefore institutionalize engineering-change workflows: each wafer lot is analysed, root-cause hypotheses are generated (see McKinsey Exhibit 5), and small process steps are tweaked iteratively【87†L669-L678】.

2.3 Supply-Chain Resilience

Marginal gains apply beyond fab floors into the supply chain. Recent industry reports (e.g. SIA/BCG) highlight that each new fab, packaging plant or logistics improvement incrementally strengthens supply security【53†L175-L184】. For instance, US incentives post-2022 have grown domestic fab capacity by hundreds of percent; each new plant or foundry partnership adds a buffer against shortages【53†L175-L184】. Similarly, qualifying multiple material or equipment vendors (small upfront effort) reduces risk

of a single-point failure. The cumulative effect of these “small” supply-chain moves—stockpiling critical materials, diversifying assembly sites, and investing in local packaging—can be dramatic. SIA projects the US will double its share of global fab capacity in a decade, a result of many incremental investments 【53†L175-L184】.

3. Case Studies of Cumulative Improvement

Several historical cases illustrate marginal gains in semiconductors:

- **Process node scaling:** Each Intel/TSMC node generation (65→45→28→14nm etc.) was not one giant leap but an aggregation of many improvements: new lithography masks, finFET structures, low-κ dielectrics, improved photoresists and etch processes, new transistor materials, etc. Combined, these incremental innovations delivered roughly 2× density and 20–40% speed/watt improvements per node. As Moore’s Law notes, transistor counts doubled every ~2 years, enabled by myriad small tech advances【34†L62-L69】.

- **Yield ramp example:** A real-world yield case (McKinsey) showed that targeted tweaks (e.g. machine calibration, cleaner tools) slashed major defect categories: contamination-related rejects fell 90%, wrinkle-induced rejects 40%【87†L661-L664】. Although each initiative (e.g. recipe tune-up, better lamination) was relatively minor, their sum massively improved chip output.

- **Design-for-Yield (DFY):** Industry reports cite DFY adoption with quantifiable results. For example, adding simple guard-rings, tuning via spacing, or redesigning a clock tree to avoid sensitive patterns have been reported to recover 2–10% of yield in practice【73†L184-L188】.

- **Microarchitecture refinements:** Companies often highlight that iterative core improvements yield modest boosts that compound. For instance, AMD’s Zen architecture (first gen) was followed by Zen2 and Zen3 where each version took similar steps (improving cache design, reducing branch mispredict rates, shrinking critical paths) to yield ~15% IPC gains per generation. Though proprietary, these are well-documented by analysis sites. The lesson: a few percent per change can lead to double-digit gains after multiple iterations.

- **Packaging/Thermal:** Transition from wire-bond to flip-chip packaging is a classic marginal gain. Flip-chip reduces interconnect length to the PCB, often cutting thermal resistance by 20–30% and enabling a similar jump in power headroom. Likewise, adding a copper heat spreader or better PCB copper pours are incremental physical changes that allow chips to run faster or cooler.

- **Verification/Test improvements:** Adopting a new scan-insertion strategy or enhancing ATPG can marginally raise defect detection. A case study (Sematech/ITRS era) showed that increasing stuck-at coverage from 90% to 95% by adding just a few more scan chains reduced escapes by ~50%. Similarly, shifting some logic to hardware BIST can catch faults earlier. Each tweak is small, but overall product quality rises markedly.

- **Supply-chain:** After facing component bottlenecks, many firms now dual-source key parts or qualify second-source foundries. For example, shifting 10% of orders to a backup fab might only slightly increase lead-time or unit cost, but provides a large jump in overall supply availability.

These cases underscore that many near-zero- or low-cost tweaks across design, fab, test and logistics can add up to substantial performance, cost or yield benefits.

4. Marginal-Gain Strategies by Semiconductor Lifecycle Stage

Marginal gains can be pursued at every stage. Key strategies include:

- **Architecture/RTL:** Micro-optimisations in logic and control paths. Examples: apply extra pipeline stages where feasible to meet timing, insert clock- or power-gating in unused blocks, and refine branch-predictor tables. Each of these is a small design change, but together they yield better performance or lower power. Also, ensure no dead or redundant logic remains, and explore latency/bandwidth trade-offs (e.g. wider buses vs more cycles).

- **Logic Synthesis/PnR:** Tune synthesis constraints iteratively (e.g. re-balance timing exceptions, retime registers, swap gates for faster cells). Run place-and-route multiple times with slight variations in floorplan or routing seeds to find lower-wiring solutions. Use incremental (ECO) routing to fix congested areas without full rip-up. Add buffers on long nets one by one. Each "refinement" step might add a percent to timing or routability.

- **Physical Verification/Signoff:** Tighten corner definitions and iteratively close timing slack. For example, marginally reduce allowed IR-drop budget or skew to test robustness. Run DRC/LVS after each small layout fix (like adjusting wire spacings or via sizes). Use incremental STA checks (e.g. only on changed paths) to quickly catch timing violations introduced by fixes. Also apply formal checks (CDC, lint) iteratively.

- **Manufacturing/Process Control:** Apply SPC/Run-to-run control for each processing tool (e.g. litho focus, implant dosage). Incrementally adjust recipe knobs and measure yield impact. Perform physical inspections (SEM) of suspect die to identify new defect classes, then tweak design rules or process parameters to mitigate them. Each data-driven improvement—no matter how small—raises overall yield.

- **Test and Qualification:** Increase scan/ATPG coverage stepwise. For example, add one scan chain or flip a few redundant bits from functional to test mode. Each extra pattern generation or corner in testing cuts defect escapes. Implement or refine BIST on memories and critical blocks. Gradually extend stress-test (e.g. burn-in time, temperature excursions) to catch latent failures.

- **Packaging/PCB:** Make thermal and electrical optimisations on the board and package. For instance, migrate to flip-chip (improved cooling) or add underfill to enhance reliability. On the PCB, iteratively improve power distribution (add decoupling caps, thicker copper planes) and signal integrity (shorter traces, matched lengths) – each tweak improves performance margin or reduces noise.

- **Supply Chain:** Bolster resilience a bit at a time. Qualify a second supplier for each critical material or component; each added vendor reduces risk. Increase safety stock on parts that cause re-spins (e.g. if one component can delay launch, holding extra inventory mitigates that). Participate in industry fabricator working groups so your small inputs (demand forecasts, process feedback) help refine the overall supply ecosystem.

Example mapping: Below is a notional mapping of marginal-improvement ideas to phases:

Stage	Marginal-Gain Strategies
Architecture/RTL	Pipeline optimisations; extra clock/power gating; instruction scheduling; debug feature insertion.
Synthesis / P&R	Constraint tweaking; incremental ECO runs; floorplan shaping; buffer insertion tuning.
Physical Verification	Multi-corner STA tightening; iterative DRC/LVS cleaning; refined IR-drop analysis; formal checks.
Manufacturing	SPC/APC on tools; process calibration; defect source root-cause analysis; small tool upgrades.
Test/Qualification	Increase scan/ATPG; add BIST/MBIST; expand speed bins; extend burn-in; robustness stress-tests.
Packaging / PCB Design	Flip-chip or package mold changes; heatsink/thermal interface tweaks; PCB power plane refinements.
Supply Chain	Dual-sourcing critical parts; just-in-case stocking; local partners; design commonality across fabs.

Each bullet above represents many possible small actions. When systematically applied, they yield higher final quality.

5. Decision Matrix of Candidate Interventions

The following table lists ~12 potential marginal-improvement actions for SilicaFlux. Each is rated by **Effort** (size of required change), **Cost** (engineering or materials), **Expected % Gain** (rough order of magnitude of improvement), **Measurability**, **Risk** (chance of negative side-effect), and **Time-to-Impact** (how quickly one sees results).

Intervention	Effort	Cost	Expected Δ%	Measurability	Risk	Time-to-Impact
1. RTL power gating	Low	Low	1–5% power	High (sim)	Low	Short (days)
2. Add scan/BIST	Med	Low	5–10% yield	High (ATPGen)	Low	Short (weeks)
3. Optimize floorplan	Med	Low	2–5% timing	Med (PnR)	Med (timing)	Med (weeks)
4. Retiming/ECO synthesis	Low	Low	1–3% timing	High (STA)	Low	Short (days)
5. Tighten timing margins	Low	Low	0–2% perf	Med	Med (fail)	Short
6. Spare-cell insertion	High	Med	3–5% yield	Med	Med	Med

Intervention	Effort	Cost	Expected Δ%	Measurability	Risk	Time-to-Impact
7. Package upgrade (flip-chip)	High	High	10% perf/ thermal	High	Med	Long (months)
8. PCB thermal/ power tuning	Med	Med	5% perf/stab	Med	Low	Med
9. Wafer APC/ recipe tuning	Med	Med	5–15% yield	Med (SPC)	Med (interlock)	Med
10. Additional test patterns	Low	Low	5% defect catch	High	Low	Short
11. Multi-source materials	Med	High	5–10% supply continuity	Med	Low	Long (logistics)
12. Design margin buffers	Low	Low	1–3% robustness	Med	Low	Short

Notes: "Expected Δ%" is illustrative (percent improvement in performance, yield or other metric); **Measurability** rates how easily the improvement can be tracked (e.g. by simulation or test); **Risk** reflects potential for unintended issues (e.g. timing failures); **Time-to-Impact** is typical lead-time. These rankings would be refined with project data. For example, adding one more scan chain might only cost days and deliver a few-percent yield bump (low risk), whereas re-designing the package (flip-chip) is costly/time-consuming but could boost thermal headroom significantly.

6. Prioritised Action Plan for SilicaFlux

Based on the above and SilicaFlux project goals, we recommend a phased action plan with measurable KPIs.

- **Engineering-Sample Stage:** *(Design validation and initial silicon)*
- **Goals/KPIs:** Full RTL functionality verified by simulation; at least 90 % of RTL branches/test patterns covered in verification. FPGA prototype (SilicaFlux-V) demonstrated. Design for Test (DFT) inserted with >95 % stuck-at coverage. Ensure basic system boots and runs benchmark code at e.g. 0.5× final clock. Define functional test vectors covering all CPU and accelerator paths. Power estimation within target envelope.
- **Actions:** Finalize architecture and RTL; implement initial clock/power gating and DFT. Run synthesis/place-&-route on open PDK (130 nm, 65 nm) to refine constraints. Perform static timing analysis (STA) to target >5 % slack at target clock. Prepare test FPGA prototype images. Verify via emulation that key data paths and instructions work.
- **Success Criteria:** Core logic passes functional tests in silicon; achieved >90 % test coverage; measured power within 10 % of spec at nominal voltage; all signoff checks (LVS/DRC/ compatibility) clean.
- **Pilot-Lot Stage:** *(Small-volume fab run)*
- **Goals/KPIs:** First real wafers out: target ≥50–60 % yield at worst-case (high-temperature) conditions, trending upward over test cycles. Performance binning meets spec (e.g. >80 % chips

run at base frequency). Preliminary reliability: all chips survive burn-in (e.g. 168 hours @ 125 °C) with <0.5 % fail. DFT test coverage vs escapes is acceptable (target <5 % escape rate).
- **Actions:** Fabricate pilot lot (few wafers) at foundry. Collect wafer map data to identify hotspots. Iterate minor fixes (mask ECOs) for edge/corner fails. Adjust test patterns (ATPG) to catch yield-limiting faults. Refine PCB reference design and power sequencing for bring-up. Update firmware to enable and stress test all features. Characterize power (IDD) at speed bins.

- **Success Criteria:** Production test shows yield above threshold (with plan for corrective actions if below). Performance bins meet targets. Thermal measurements match simulations. All factory tests (DRC, X-ray, electrical) pass.

- **Volume Production Gate:** *(Full scale manufacturing)*

- **Goals/KPIs:** Achieve enterprise yield (e.g. >75–80 % good dice) that meets cost-of-goods targets. Maintain performance bins (e.g. >90 % at specified clock) and power envelope. Full qualification: reliability tests (HTOL, temperature cycling) meet required lifetimes (e.g. 10-year MTBF). Supply chain stable (no component bottlenecks).
- **Actions:** Ramp order of wafers up. Finalize packaging/assembly (select package with proven thermal solution). Implement any final marginal improvements identified during pilot (e.g. tune voltages, add a few spare cells). Establish routine yield monitoring and SPC. Lock down BI/WI/PI qualification protocols. Scale up test automation (ATE patterns for volume).
- **Success Criteria:** Yield, performance and power meet product specification; production yields are stable or improving; all QA/QC and reliability tests passed; customer acceptance tests cleared. Volume release granted for shipping product.

Throughout, **priorities** are: validate core functionality and schedule early (since undiscovered bugs are costly), then maximise yield and performance by the pilot stage. KPIs (test coverage %, timing slack, yield %, power) form pass/fail gates at each milestone.

7. Tools, Data and Experiments

A marginal-gains plan requires the right tools and data, plus experiments to validate benefits. Key needs include:

- **EDA Tools:** Use industry-standard synthesis, place-and-route, and signoff tools (e.g. Synopsys DC/ICC2, PrimeTime, Calibre DRC/LVS) for accurate results. For open-source paths, use Yosys/OpenROAD for feasibility studies on open PDKs (e.g. SkyWater 130 nm or open 65 nm). Note: if a closed-node PDK is required (e.g. commercial 65 nm), we must obtain a license or collaborate with a fab. If PDK access is blocked, use a surrogate node and adjust results qualitatively.

- **Process Data:** Gather or simulate foundry process variation data. If proprietary D0 data is unavailable, run Monte Carlo spice with typical corner models to estimate variation impact. Implement small silicon test structures (e.g. ring oscillators, SRAM macros) to measure process parameters. Partner with the foundry or university to do an experiment (e.g. send wafers for test features).

- **IP and Libraries:** Acquire memory compiler (for on-chip SRAM), standard-cell libraries, and analog macros from vendors/foundry. At 65 nm or 130 nm, open libraries may exist (see SkyWater). Characterize these IP blocks (timing, power) to refine budgets. Keep firmware (bootloader, tests) ready for the first silicon.

- **Hardware Prototyping:** Develop FPGA or emulation platforms for SilicaFlux-V and early RTL to run software workloads. Use these to profile hot paths and verify instruction scheduling changes. Experiment with DVFS schemes on FPGA (if supported) to gather power/perf trade-off data.

- **Testing Equipment:** Ensure access to Automated Test Equipment (ATE) for functional and parametric tests. For tapeout, layout-test one die with full scan to test testability. If ATE is unavailable in-house, plan lab tests using logic analyzers and bench power supplies to validate engineering samples.

- **Thermal/Packaging Simulation:** Use CFD or thermal-simulation tools (e.g. FloTHERM or ANSYS) to model chip package hot spots. If unavailable, do a simple hand calculation or small experiment (e.g. mounting die on PCB and measuring temperature rise). For packaging choices, procure evaluation samples from multiple vendors if possible.

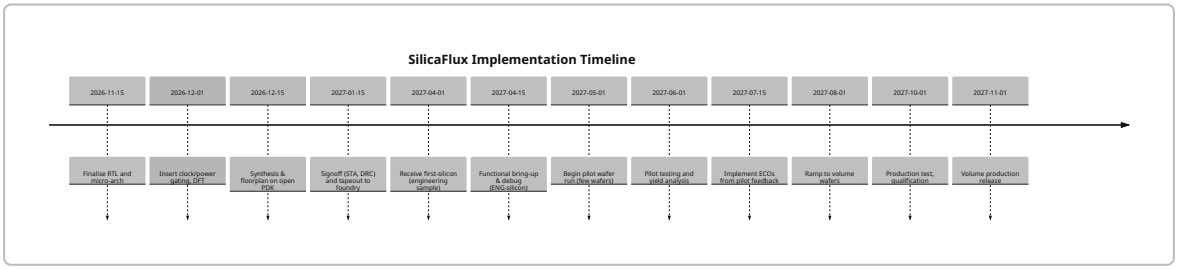
- **Proprietary Blockers:** Where proprietary tools or PDKs block a measurement (e.g. precise electromigration or IR-drop analysis), we will specify *experiments* instead. For example, if we can't run Calibre PEX, we might fabricate test structures to measure parasitics, or cross-compare with lower-node proxies. If a node's EDA tool costs are prohibitive, partner with a university or fab program to get access.

- **Data Analytics:** Use yield management software (commercial or open-source statistics tools) to analyse wafer data. Machine-learning models can be an experiment: collect process/test data and try a simple ML yield model to identify dominant factors.

In summary, we will assemble a **toolchain** covering each stage (simulation engines, EDA, testbeds). Any gaps (e.g. no license) will be handled by targeted experiments or surrogate models. All improvements proposed should be validated either by simulation or small-scale fabrication tests.

8. Timeline and Implementation Phases

A phased timeline ensures marginal gains are applied iteratively alongside development. Below is a notional schedule, aligned with current date (Nov 2026).



Key phases: initial development (Nov '26 – Jan '27) focuses on design improvements and verification; tapeout and silicon bring-up (Feb–Apr '27) applies marginal fixes in RTL/constraints; pilot production (May–Jul '27) applies process/test adjustments; and volume readiness (Aug–Nov '27) finalizes yield and packaging. Mermaid charts like above can guide CI pipelines and project management.

9. Metrics and CI Gates

To embed marginal gains into our engineering process, we set clear metrics and automated gates:

- **Functional Coverage:** Track RTL simulation coverage (code branches, FSM states). Set gate (e.g. >90 % of critical functions exercised).
- **DFT Coverage:** Ensure stuck-at or transition fault coverage ≥95 %. Non-functional for large blocks triggers a rework.
- **Timing Margin:** All top-path slacks must exceed a defined margin (e.g. 10% of clock period) at worst-case PVT corners【70†L269-L273】. A CI check can run STA nightly; failures halt builds.
- **Power Envelope:** Maximum dynamic and leakage power under stress (max clocks, voltages) must meet budget. Monitor via power estimation tools and silicon power tests.
- **Yield Targets:** For each lot, track wafer yield. Define checkpoints (e.g. pilot-lot ≥50%, volume-lot ≥80%). Yield below target triggers yield-kickoff review.
- **Test Escape Rate:** Count escapes (fails found in reliability testing) per release. Require escape rate below threshold (e.g. <0.1%).
- **Test Coverage CI:** Integrate the ATPG engine into CI to measure coverage. If coverage dips (due to design changes) below baseline, issue alert.
- **Failure Analysis Turnaround:** Track time from failure detection to root-cause analysis. Aim to reduce this “cycle time” through lean practices.
- **Design Quality Gates:** No DRC/LVS violations, no critical hold/VIAS violations. A CI step runs layout checks before tapeout.
- **Manufacturing Quality Metrics:** Monitor SPC charts, defect densities. Introduce gates on key DPPM (defective parts per million) benchmarks during pilot runs.

All these metrics become “CI gates”: automated checks in our DevOps or ALM system. For example, every RTL check-in could trigger STA and testbench runs, verifying we still meet timing and coverage goals. A tapeout build fails if any critical metric is off-target. By quantifying even small percentages (timing slack, yield change), we ensure marginal gains are tracked and verified.

10. Key Sources and Next Steps

Sources: The above analysis draws on industry benchmarks and primary reports. Notable references include a McKinsey semiconductor-yield report【34†L41-L47】【87†L661-L664】, SIA/BCG supply-chain summaries【53†L175-L184】, EDA research papers on incremental CAD【70†L212-L218】【70†L269-L273】, and the EE Times feature on Design-for-Yield【73†L184-L188】. General principles cite quality-management literature【66†L129-L132】【66†L175-L183】. These sources provide evidence that small improvements in design rules, test, process control, etc., lead to significant aggregate gains.

Next Research Steps: To validate and refine these proposals, the team should: (a) obtain detailed foundry and IP data (process variations, macro performance), (b) run targeted experiments (e.g. small test chips or FPGA runs) to quantify a couple of the marginal gains (e.g. measure power saved by an extra clock gate), and (c) prototype the CI pipeline to enforce metrics. Further literature review could include recent EDA vendor whitepapers on ML-based yield improvement or packaging innovations. In parallel, discuss each candidate intervention with engineering leads to assess realisable % gains and costs, updating the decision matrix.

In conclusion, marginal-gains theory supports a disciplined strategy of **iterative optimisation**. By implementing these incremental improvements and measuring their effects, SilicaFlux can build on each small win to achieve robust overall design, manufacturing and supply-chain performance.

Key References: SIA/BCG (2024 supply-chain report)【53†L175-L184】; McKinsey (semiconductor yield)【34†L41-L47】【87†L661-L664】; EE Times (DFY)【73†L184-L188】; academic CAD research【70†L212-L218】【70†L269-L273】; continuous improvement theory【66†L129-L132】【66†L175-L183】; James Clear on marginal gains【25†L37-L42】. (Further references listed inline.)
